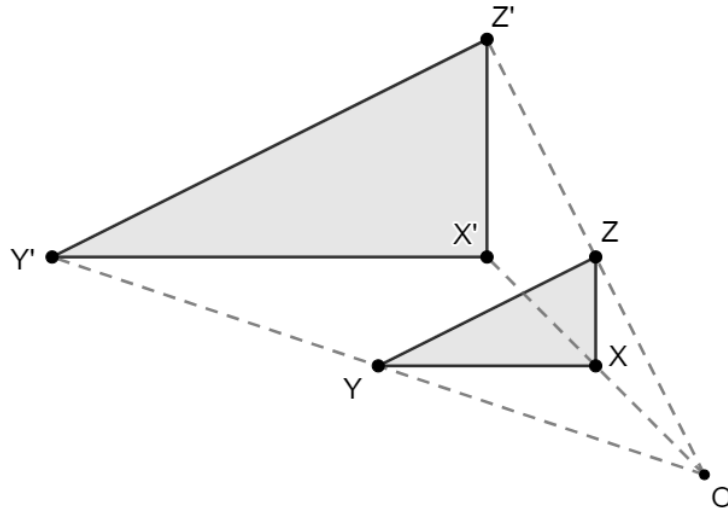


Geometry
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$\triangle X'Y'Z'$ was created by dilating $\triangle XYZ$. The length of $\overline{XY}$ is 2 and the length of $\overline{X'Y'}$ is 4.



1. $\triangle XYZ$ is the

☐ image
☒ preimage

, $\triangle X'Y'Z'$ is the

☒ image
☐ preimage

, and the center of dilation is point

☐ X
☐ X'
☒ O

.

2. Determine the dilation ratio.

$$\frac{4}{2} = 2$$

3. If the length of $\overline{Y'Z'}$ is $2\sqrt{5}$, determine the length of $\overline{YZ}$.

$$\frac{2\sqrt{5}}{2} = \sqrt{5}$$

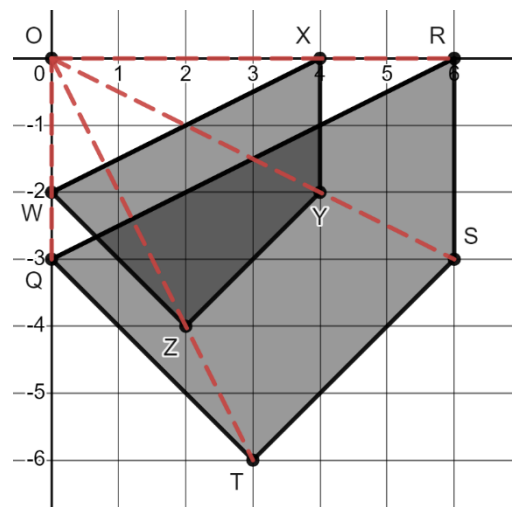
4. What is the ratio of $\frac{\overline{XZ}}{\overline{X'Z'}}$?

$$\frac{1}{2}$$

5. What is the ratio of $\frac{\overline{YY'}}{\overline{OY}}$?

$$\frac{1}{1}$$

WXYZ and QRST are shown on the coordinate grid, where WXYZ is a dilation of QRST.



6. Complete the table:

Segment	Slope	Distance
$\overline{YZ}$	$\frac{2}{2} = 1$	$2\sqrt{2}$
$\overline{ST}$	$\frac{3}{3} = 1$	$3\sqrt{2}$
$\overline{WX}$	$\frac{2}{4} = \frac{1}{2}$	$2\sqrt{5}$
$\overline{QR}$	$\frac{3}{6} = \frac{1}{2}$	$3\sqrt{5}$

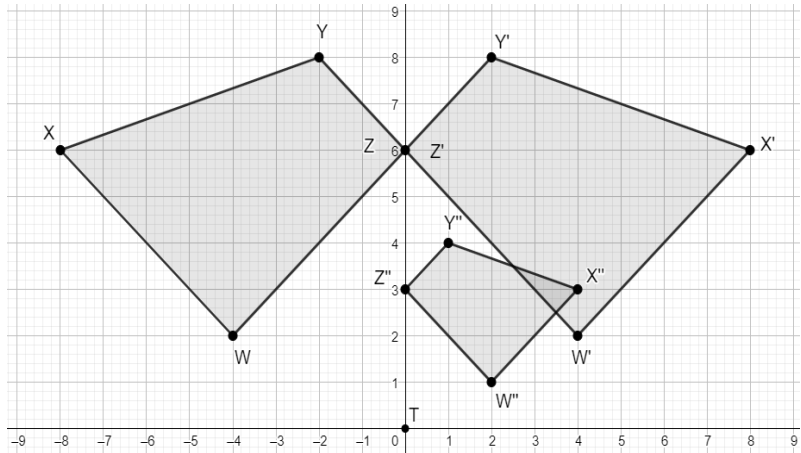
7. Show the dilation on the coordinate grid by drawing dotted lines from the center of dilation through each vertex.
Shown on coordinate grid.

8. Write a description of the dilation by giving the center of dilation and the scale factor.
QRST was dilated from the origin by a scale factor of $\frac{2}{3}$ to result in the image WXYZ.

9. What is the ratio $\frac{\overline{OX}}{\overline{OR}}$?
 $\frac{4}{6} = \frac{2}{3}$

10. What is the ratio $\frac{\overline{OX}}{\overline{XR}}$?
 $\frac{4}{2} = 2$

A sequence of transformations of WXYZ is shown on the coordinate grid. WXYZ is reflected then dilated, where W'X'Y'Z' is the reflection and W''X''Y''Z'' is the dilation of W'X'Y'Z'.



Complete the descriptions:

- 1. WXYZ was reflected over the y-axis to create W'X'Y'Z'.
- 2. W'X'Y'Z' was dilated using a scale factor of 1/2 centered at T to create W''X''Y''Z''.
- 3. Complete the table:

Transformation	Algebraic Description
Reflection	$(x,y) \rightarrow (-x,y)$
Dilation	$(x,y) \rightarrow (\frac{1}{2}x,\frac{1}{2}y)$

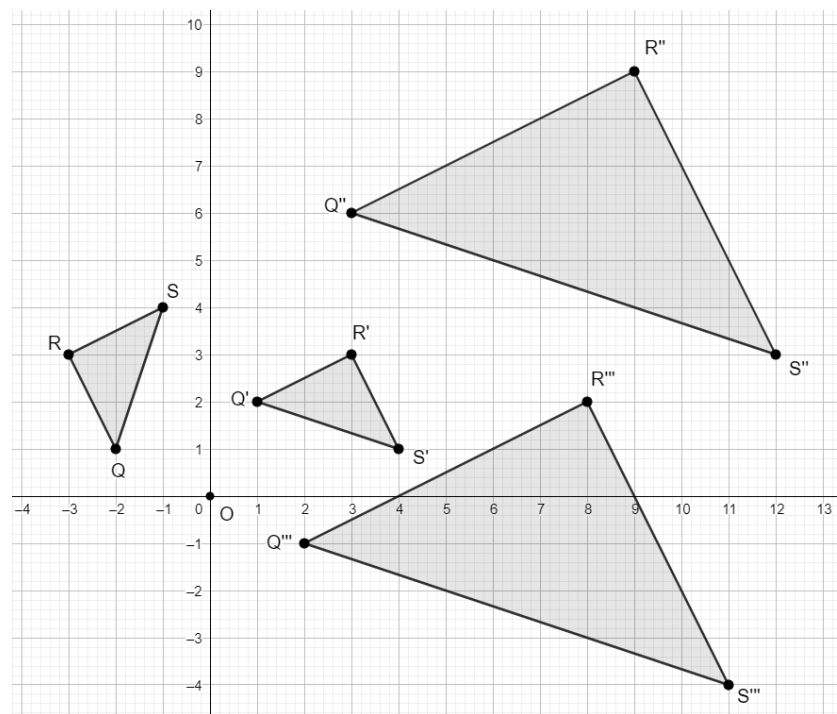
- 4. Determine the coordinates if WXYZ is first dilated then reflected:

Preimage WXYZ		Dilation W'X'Y'Z'		Reflection W''X''Y''Z''	
W	(-4,2)	W'	(-2,1)	W''	(2,1)
X	(-8,6)	X'	(-4,3)	X''	(4,3)
Y	(-2,8)	Y'	(-1,4)	Y''	(1,4)
Z	(0,6)	Z'	(0,3)	Z''	(0,3)

- 5. Based on the results above, the sequence of transformations commutative.

✗ is
o is not

$QRS, Q'R'S', Q''R''S'',$ and $Q'''R'''S'''$ are shown on the coordinate grid.



6. Determine what transformations occurred and in what order:
- QRS is first

☐ translated

 to $Q'R'S'$, then

☐ translated

 to $Q''R''S''$,

☐ reflected

☒ rotated

☐ dilated

☐ reflected

☐ rotated

☒ dilated
- and finally

☒ translated

 to $Q'''R'''S'''$.

☐ reflected

☐ rotated

☐ dilated
7. Describe the transformation for $QRS \rightarrow Q'R'S'$.
Rotation 90° clockwise about the origin
8. Describe the transformation for $Q'R'S' \rightarrow Q''R''S''$.
Dilation by a scale factor of 3 from the origin
9. Describe the transformation for $Q''R''S'' \rightarrow Q'''R'''S'''$.
Translation 1 unit left and 7 units down
10. Write the algebraic description for the sequence of transformations from 7-9:

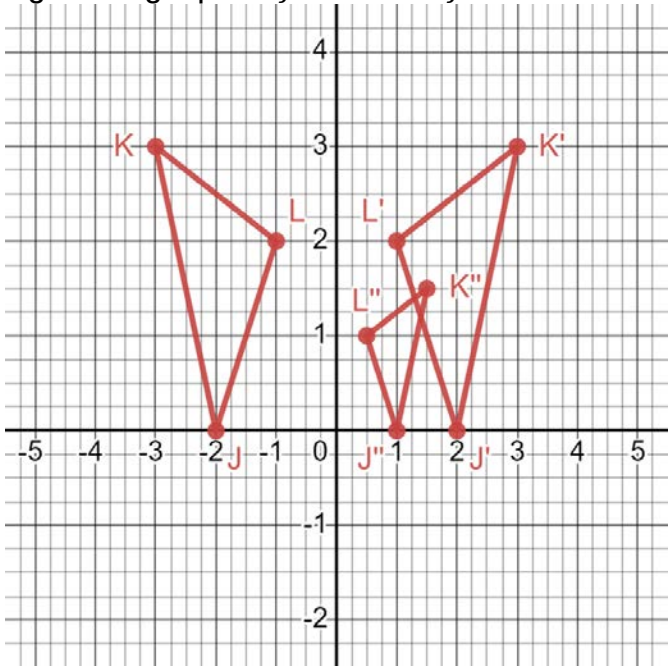
Transformation	Algebraic Description
Rotation	$(x,y) \rightarrow (y,-x)$
Dilation	$(x,y) \rightarrow (3x,3y)$
Translation	$(x,y) \rightarrow (x-1,y-7)$

The coordinates of $\triangle JKL$ and $\triangle J'K'L'$ are shown in the table, where $\triangle JKL$ is the preimage:

$\triangle JKL$		
Vertex	x	y
J	-2	0
K	-3	3
L	-1	2

$\triangle J'K'L'$		
Vertex	x	y
J'	2	0
K'	3	3
L'	1	2

1. Use the coordinate grid to graph $\triangle JKL$ and $\triangle J'K'L'$.



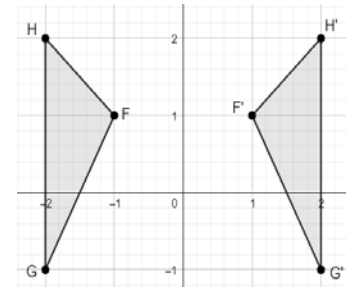
- What type of transformation was applied to the preimage $\triangle JKL$?
 Reflection across the y -axis
- Was distance preserved in this transformation?
 Yes
- $\triangle J'K'L'$ is further transformed by dilating with a scale factor of $\frac{1}{2}$ centered at the origin to create $\triangle J''K''L''$. Graph $\triangle J''K''L''$ on the coordinate grid above.
 Shown on coordinate grid
- Was distance preserved in this second transformation from $\triangle J'K'L'$ to $\triangle J''K''L''$?
 No

Complete the following statements about the transformation of preimage FGH :

6. Triangle $F'G'H'$ was created by a

- ☐ translation.
- ☒ reflection.
- ☐ rotation.
- ☐ dilation.

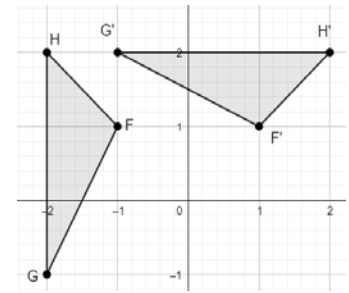
This transformation ☒ does ☐ does not preserve distance.



7. Triangle $F'G'H'$ was created by a

- ☐ translation.
- ☐ reflection.
- ☒ rotation.
- ☐ dilation.

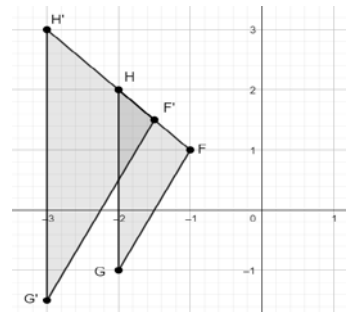
This transformation ☒ does ☐ does not preserve distance.



8. Triangle $F'G'H'$ was created by a

- ☐ translation.
- ☐ reflection.
- ☐ rotation.
- ☒ dilation.

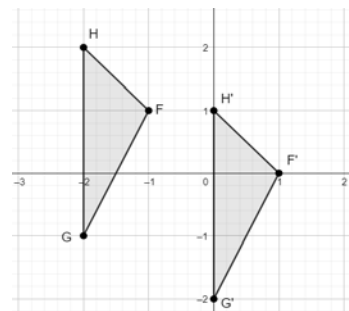
This transformation ☐ does ☒ does not preserve distance.



9. Triangle $F'G'H'$ was created by a

- ☒ translation.
- ☐ reflection.
- ☐ rotation.
- ☐ dilation.

This transformation ☒ does ☐ does not preserve distance.

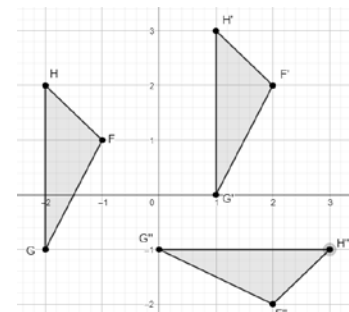


10. Triangle $F''G''H''$ was created by a transformation sequence of first a

- ☐ translation.
- ☐ reflection.
- ☒ rotation.
- ☐ dilation.

- ☒ translation
- ☐ reflection
- ☐ rotation
- ☐ dilation

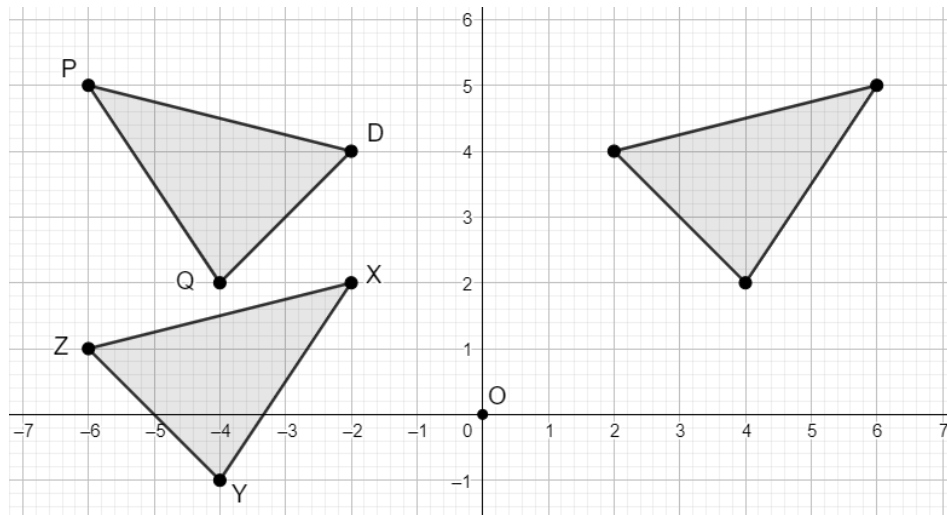
This transformation sequence ☒ does ☐ does not preserve distance.



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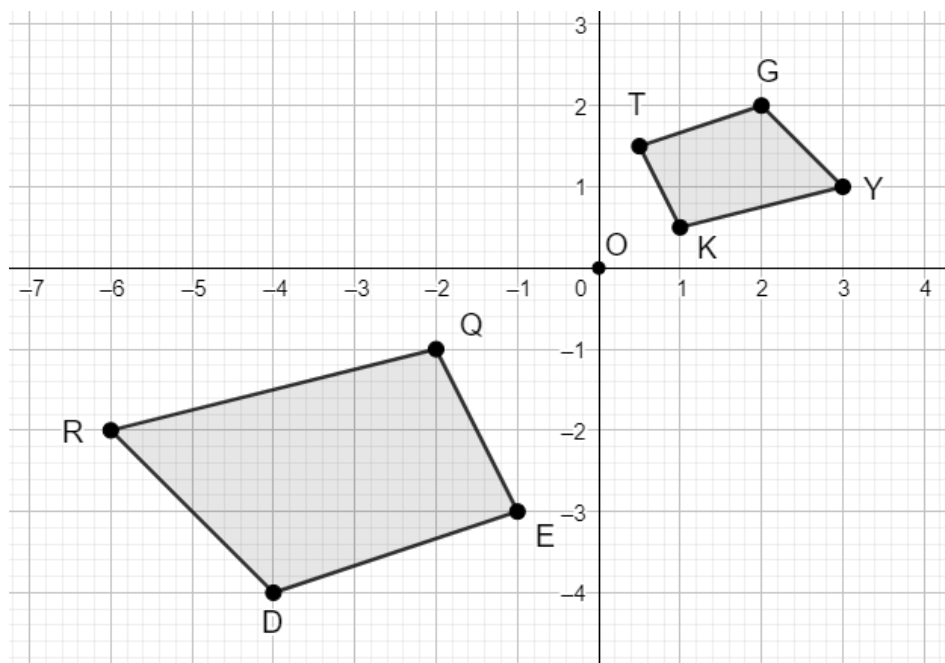
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$\triangle XYZ$ is the result of a sequence of transformations of $\triangle PDQ$. The unlabelled triangle is the first transformation.



- Describe the first transformation of $\triangle PDQ$.
The first transformation is a reflection across the y-axis.
- Write the transformation algebraically.
 $(x, y) \rightarrow (-x, y)$
- Describe the second transformation that results in $\triangle XYZ$.
The second transformation is a translation 8 units left and 3 units down.
- Write the second transformation algebraically.
 $(x, y) \rightarrow (x - 8, y - 3)$
- Explain if the sequence of transformations preserves distance.
Yes, the sequence preserves distance because reflections and translations preserve distance.

KTGY is the result of a sequence of transformations of QEDR.



Ian and Natalie were determining the sequence of transformations, but each came up with a different sequence. Complete the table for the missing portions of their transformations.

			First Transformation	Second Transformation
6.	Ian's Sequence	Written Description	Rotate 180° about the origin.	$(x,y) \rightarrow (-x,-y)$
7.		Algebraic Description	Dilate using a scale factor of $\frac{1}{2}$ centered at the origin.	$(x,y) \rightarrow \left(\frac{1}{2}x,\frac{1}{2}y\right)$
8.	Natalie's Sequence	Written Description	Dilate using a scale factor of $\frac{1}{2}$ centered at the origin.	$(x,y) \rightarrow \left(\frac{1}{2}x,\frac{1}{2}y\right)$
9.		Algebraic Description	Rotate 180° about the origin.	$(x,y) \rightarrow (-x,-y)$

10. Considering both proposed sequences, you can conclude that

- ☐ Ian's sequence is

☐ Natalie's sequence is

☒ both sequences are

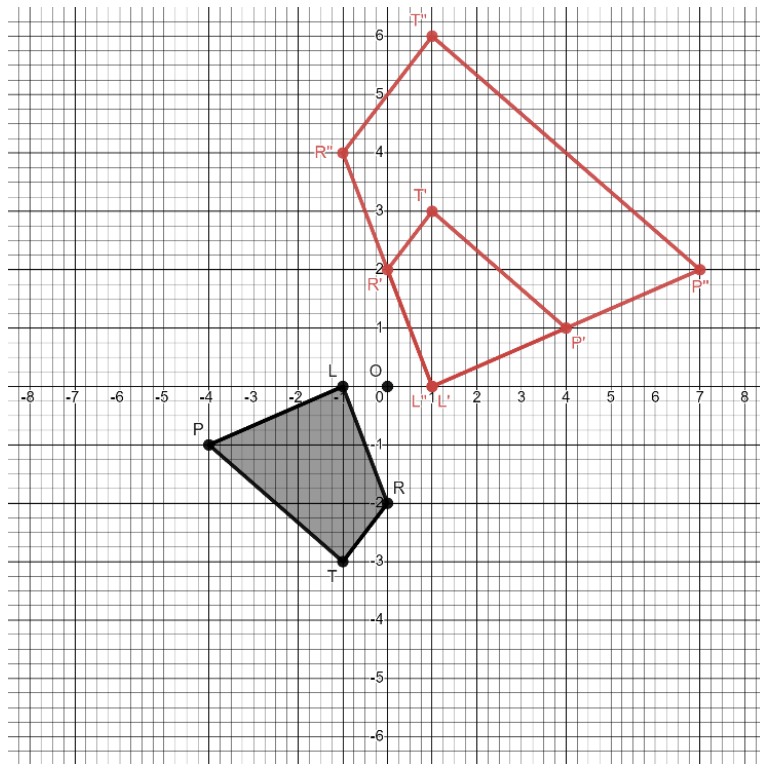
correct.

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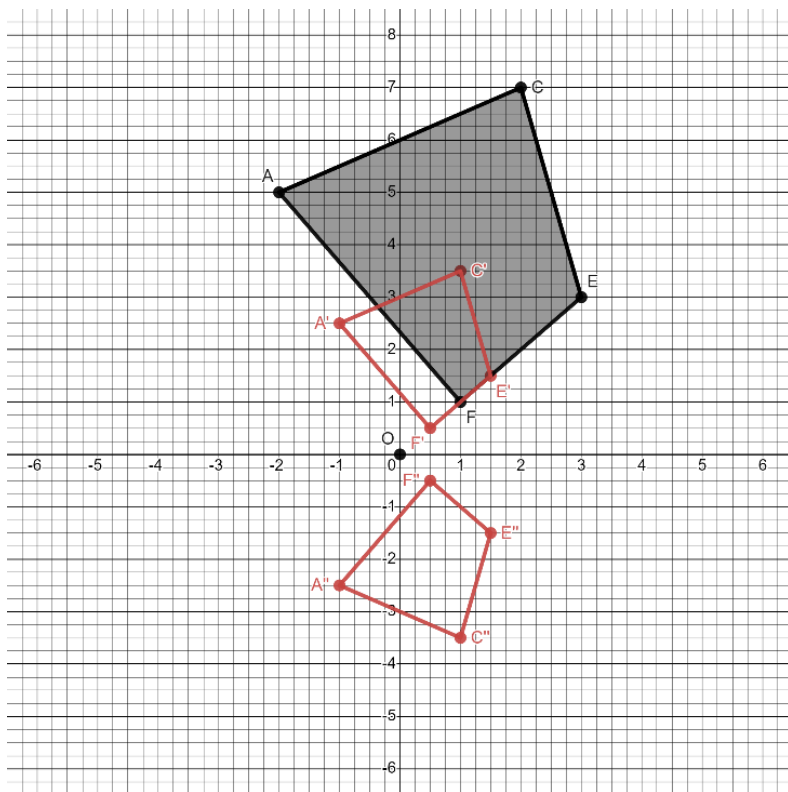
$L RTP$ and point O are shown on the coordinate plane.

1. Draw the quadrilateral that is a 180° rotation of $L RTP$ about the origin. Label the quadrilateral $L' R' T' P'$.
2. Draw the quadrilateral that is a dilation of $L' R' T' P'$ with a center of dilation at point L' and a scale factor of 2. Label the quadrilateral $L'' R'' T'' P''$.



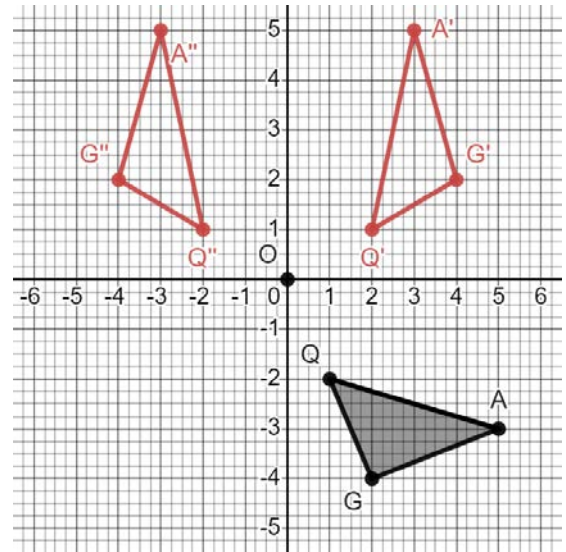
$ACEF$ and point O are shown on the coordinate plane.

3. Draw the quadrilateral that is a dilation of $ACEF$ with a center of dilation at point O and a scale factor of $\frac{1}{2}$. Label the quadrilateral $A' C' E' F'$.
4. Draw the quadrilateral that is a reflection of $A' C' E' F'$ across the x -axis. Label the quadrilateral $A'' C'' E'' F''$.



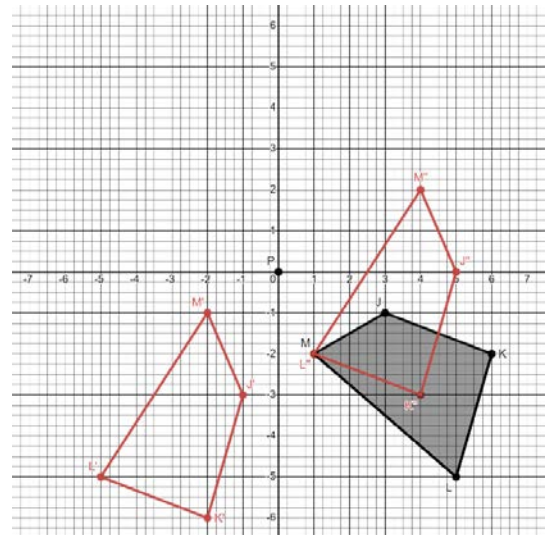
$\triangle QAG$ and point O are shown on the coordinate plane.

5. Draw the triangle that is a 90° counterclockwise rotation of $\triangle QAG$ about point O . Label the quadrilateral $\triangle Q'A'G'$.
6. Draw the triangle that is a reflection of $\triangle Q'A'G'$ across the y -axis. Label the quadrilateral $\triangle Q''A''G''$.



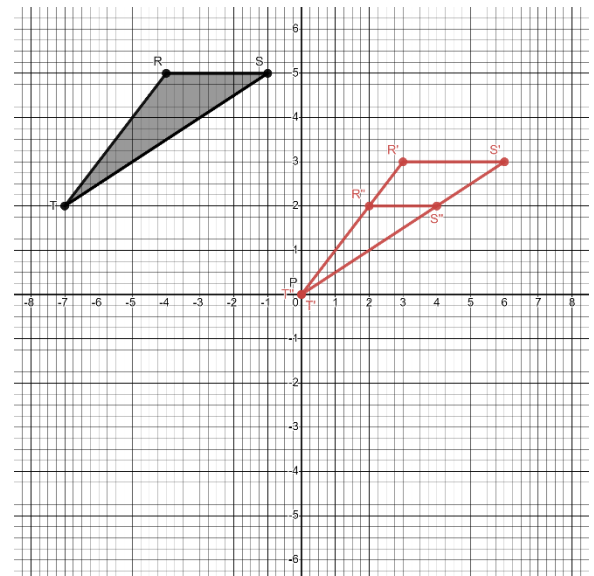
$JKLM$ and point P are shown on the coordinate plane.

7. Draw the quadrilateral that is a 90° clockwise rotation of $JKLM$ about the origin. Label the quadrilateral $J'K'L'M'$.
8. Draw the quadrilateral that is a translation of $J'K'L'M'$ 6 units right and 3 units up. Label the quadrilateral $J''K''L''M''$.



$\triangle RST$ and point P are shown on the coordinate plane.

9. Draw $\triangle R'S'T'$ as a result of $(x, y) \rightarrow (x + 7, y - 2)$.
10. Draw $\triangle R''S''T''$ as a result of $(x, y) \rightarrow \left(\frac{2}{3}x, \frac{2}{3}y\right)$.



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1.

- ☐ Only dilations
- ☐ Only translations, reflections, and rotations

✗ All transformations

corresponding angles.

result in an image with congruent

- ☐ Only dilations
- ✗ Only translations, reflections, and rotations
- ☐ All transformations

corresponding angles and sides.

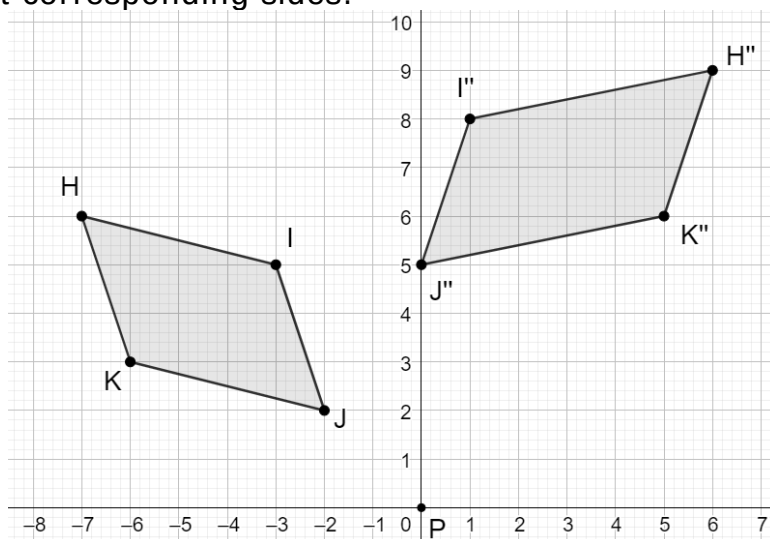
result in an image with congruent

✗ Only dilations

- ☐ Only translations, reflections, and rotations
- ☐ All transformations

but not congruent corresponding sides.

result in an image with proportional,



2. Write a sequence of transformations for preimage $HIJK$ that would show whether the figures are similar.

Figures are not similar.

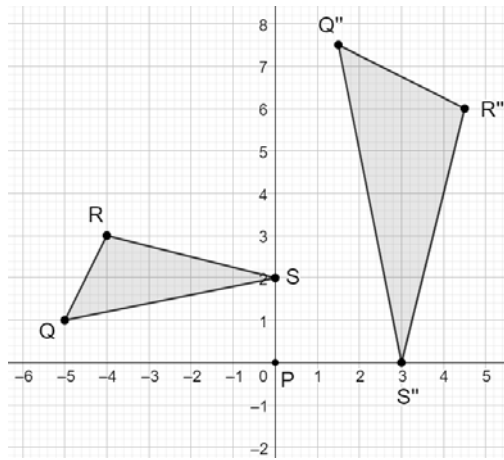
3. Is $H'I'J'K'$ similar to the preimage?

☐ Yes

✗ No

4. Explain the reason for your answer for #3.

Answers may vary. Not all of the points were transformed correctly. K' should be at (4,6) and H' should be at (5,9) in order to show the correct transformations preserving congruence and similarity from quadrilateral to quadrilateral.



5. Write a sequence of transformations for preimage $\triangle QRS$ that would show whether the figures are similar.

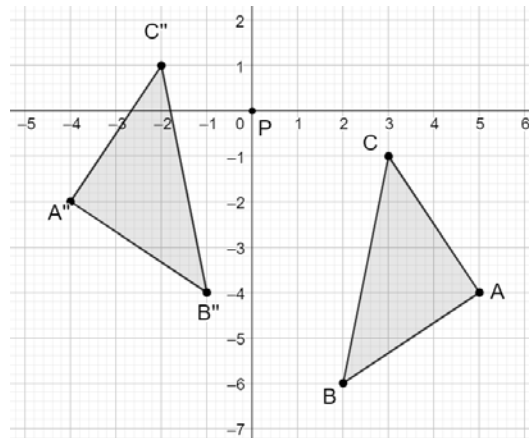
Rotation 90° clockwise about the origin, then dilation from the origin with a scale factor of $\frac{3}{2}$.

6. Is $\triangle Q''R''S''$ similar to the preimage?

☒ Yes
☐ No

7. Explain the reason for your answer for #6.

Because $\triangle Q''R''S''$ was obtained by a sequence of transformations including a dilation.



8. Write a sequence of transformations for preimage $\triangle ABC$ that would show whether the figures are similar.

Reflection across the y -axis, then translation 1 unit right and 2 units up.

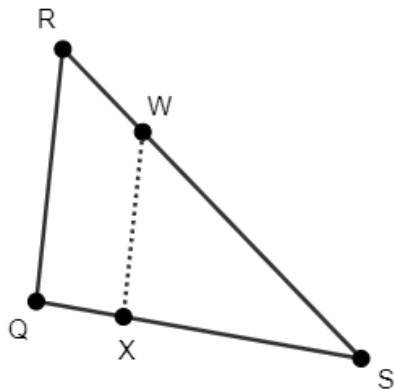
9. Is $\triangle A''B''C''$ similar to the preimage?

☒ Yes
☐ No

10. Explain the reason for your answer for #9.

Congruent figures are also similar, as you can use a dilation with a scale factor of 1.

$\triangle QRS$ is shown, where $m\angle QRS = 50^\circ$, $m\angle RQS = 94^\circ$, and $m\angle RSQ = 36^\circ$. Point W is on $\overline{RS}$ and Point X is on $\overline{QS}$. $\overline{WX}$ is drawn as a dotted line.



1. Use a ruler and a protractor to complete the table.

Measurements of $\triangle QRS$		Measurements of $\triangle XWS$	
$QR = 3.3 \text{ cm}$	$m\angle QRS = 50^\circ$	$XW = 2.4 \text{ cm}$	$m\angle XWS = 50^\circ$
$RS = 5.8 \text{ cm}$	$m\angle RQS = 94^\circ$	$WS = 4.2 \text{ cm}$	$m\angle WXS = 94^\circ$
$QS = 4.4 \text{ cm}$	$m\angle RSQ = 36^\circ$	$XS = 3.2 \text{ cm}$	$m\angle WSX = 36^\circ$

2. The corresponding angles of $\triangle QRS$ and $\triangle XWS$ are **congruent**.

3. Complete the table.

Sides	Ratio
Determine the ratio of $\overline{QR}$ to $\overline{XW}$.	11:8
Determine the ratio of $\overline{RS}$ to $\overline{WS}$.	11:8
Determine the ratio of $\overline{QS}$ to $\overline{XS}$.	11:8

4. The corresponding sides of $\triangle QRS$ and $\triangle XWS$ are **proportional**.

5. $\triangle QRS$ is similar to $\triangle XWS$.
6. Describe a sequence of transformations that can be used to map $\triangle QRS$ onto $\triangle XWS$.
Dilate $\triangle QRS$ by a scale factor of $\frac{8}{11}$ about Point S .

Complete each relationship for $\triangle QRS$ and $\triangle XWS$.

7. $\frac{RS}{WS} = \frac{QS}{XS}$

8. $\frac{WX}{RQ} = \frac{XS}{QS}$

9. $\frac{RQ}{WX} = \frac{RS}{WS}$

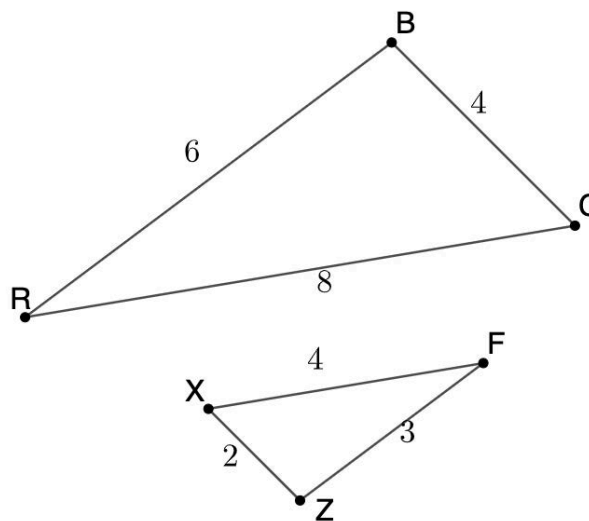
10. Observing that $\frac{RQ}{WX} \neq \frac{RW}{WS}$, your classmate Charlee claims that $\triangle QRS$ and $\triangle XWS$ are not similar. If her logic is faulty, explain why.

$\triangle QRS$ and $\triangle XWS$ are similar because side $\overline{RS}$ corresponds to side $\overline{WS}$. Charlee should have used the equation $\frac{RQ}{WX} = \frac{RS}{WS}$ to determine triangle similarity.

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$\triangle RBC$ and $\triangle XFZ$ are shown.



1. Complete the table.

Measurements of $\triangle RBC$	Measurements of $\triangle XFZ$	Ratio
$RB = 6$	$ZF = 3$	$\frac{RB}{ZF} = \frac{6}{3} = 2$
$BC = 4$	$XZ = 2$	$\frac{BC}{XZ} = \frac{4}{2} = 2$
$CR = 8$	$XF = 4$	$\frac{CR}{XF} = \frac{8}{4} = 2$

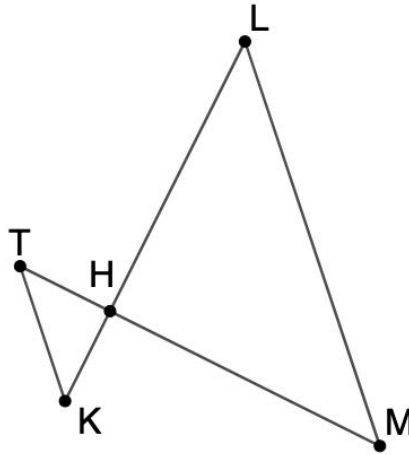
2. What similarity theorem can be applied to show that the triangles are similar?
Side-Side-Side Similarity
3. Write the similarity statement.
 $\triangle RBC \sim \triangle FZX$
4. Complete each.

$$\frac{RC}{CB} = \frac{FX}{XZ}$$

$$\frac{XF}{FZ} = \frac{CR}{RB}$$

$$\frac{BC}{RB} = \frac{ZX}{FZ}$$

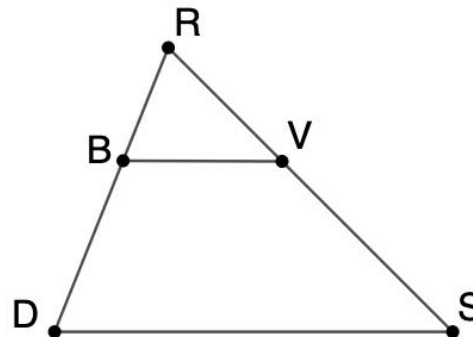
$\triangle TKH$ and $\triangle HLM$ are shown where $\overline{TK} \parallel \overline{LM}$.



Complete the two-column proof.

	Statement	Reason
	$\overline{TK} \parallel \overline{LM}$	Given
5.	$\angle HLM \cong \angle HKT$ and $\angle HML \cong \angle HTK$	If two parallel lines are intersected by a transversal, alternate interior angles are congruent.
6.	$\triangle TKH \sim \triangle MLH$	AA ~

$\triangle DRS$ and $\triangle RVB$ are shown.



Given: $RB = \frac{2}{5}RD$ and $RV = \frac{2}{5}RS$

Prove: $\triangle DRS \sim \triangle BRV$

	Statement	Reason
	$RB = \frac{2}{5}RD$ and $RV = \frac{2}{5}RS$	Given
7.	$\frac{RB}{RD} = \frac{2}{5}$ and $\frac{RV}{RS} = \frac{2}{5}$	Division Property of Equality
8.	$\frac{RB}{RD} = \frac{RV}{RS}$	Transitive Property of Equality
9.	$\angle DRS \cong \angle BRV$	Reflexive Property of Congruence
10.	$\triangle DRS \sim \triangle BRV$	SAS ~

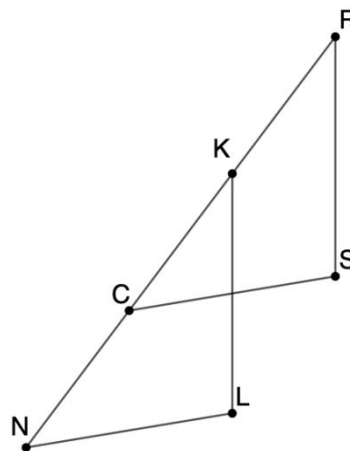
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Given: $\overline{NL} \parallel \overline{CS}$ and $\angle KRS$ and $\angle LKR$ are supplementary

Prove: $\triangle NKL \sim \triangle CRS$

Complete the two-column proof with the provided statements and reasons.



	Statement	Reason
	$\overline{NL} \parallel \overline{CS}$	Given
1.	$\angle KNL \cong \angle RCS$	If two parallel lines are intersected by a transversal, then corresponding angles are congruent.
2.	$\angle KRS$ and $\angle LKR$ are supplementary angles	Given
3.	$\overline{KL} \parallel \overline{RS}$	If consecutive interior angles are supplementary, then the two lines intersected by a transversal are parallel.
4.	$\angle NKL \cong \angle CRS$	If two parallel lines are intersected by a transversal, then corresponding angles are congruent.
5.	$\triangle NKL \sim \triangle CRS$	AA ~

$\triangle NKL \sim \triangle CRS$

AA ~

$\angle KRS$ and $\angle LKR$ are supplementary angles

If two parallel lines are intersected by a transversal, then corresponding angles are congruent.

$\angle NKL \cong \angle CRS$

If consecutive interior angles are supplementary, then two lines intersected by a transversal are parallel.

$\overline{KL} \parallel \overline{RS}$

Reflexive Property of Congruence

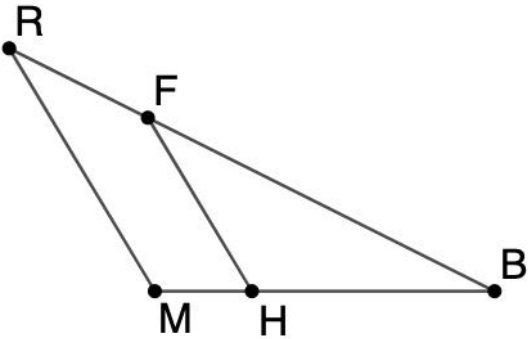
$\angle KNL \cong \angle RCS$

Given

Given: $RF = \frac{2}{7}RB$ and $MH = \frac{2}{7}MB$

Prove: $\triangle RBM \sim \triangle FBH$

Complete the two-column proof with the provided statements and reasons.



	Statement	Reason
6.	$RF = \frac{2}{7}RB$ and $MH = \frac{2}{7}MB$	Given
7.	$\frac{RF}{RB} = \frac{2}{7}$ and $\frac{MH}{MB} = \frac{2}{7}$	Division Property of Equality
8.	$\frac{RF}{RB} = \frac{MH}{MB}$	Substitution Property of Equality
9.	$\angle RBM \cong \angle FBH$	Reflexive Property of Congruence
10.	$\triangle RBM \sim \triangle FBH$	SAS ~

$\frac{RF}{RB} = \frac{2}{7}$ and $\frac{MH}{MB} = \frac{2}{7}$

$\triangle RBM \sim \triangle FBH$

$\frac{RF}{RB} = \frac{MH}{MB}$

$RF = \frac{2}{7}RB$ and $MH = \frac{2}{7}MB$

$\angle RBM \cong \angle FBH$

SAS ~

Given

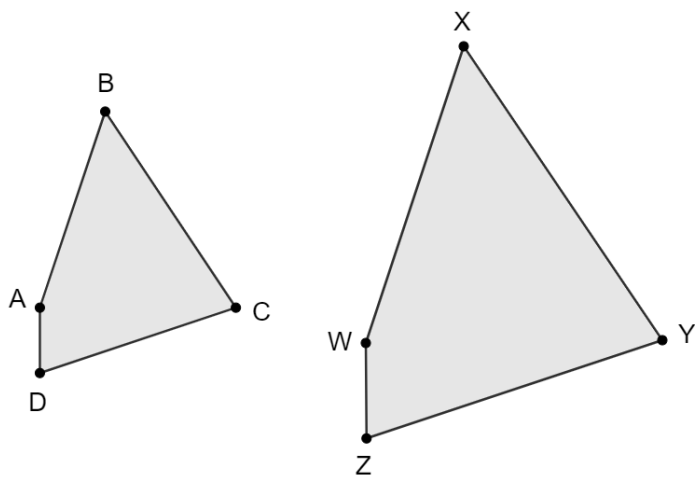
Reflexive Property of Congruence

Substitution Property of Equality

Division Property of Equality

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Fill in the table given that $ABCD \sim WXYZ$.

	Corresponding Angles	Corresponding Sides
1.	$\angle DAB \cong \angle ZWX$	$\frac{AB}{WX} = \frac{DC}{ZY}$
2.	$\angle WZY \cong \angle ADC$	$\frac{AD}{WZ} = \frac{BC}{XY}$
3.	$\angle XYZ \cong \angle BCD$	$\frac{BC}{XY} = \frac{AB}{WX}$
4.	$\angle ABC \cong \angle WXY$	$\frac{DC}{ZY} = \frac{AD}{WZ}$

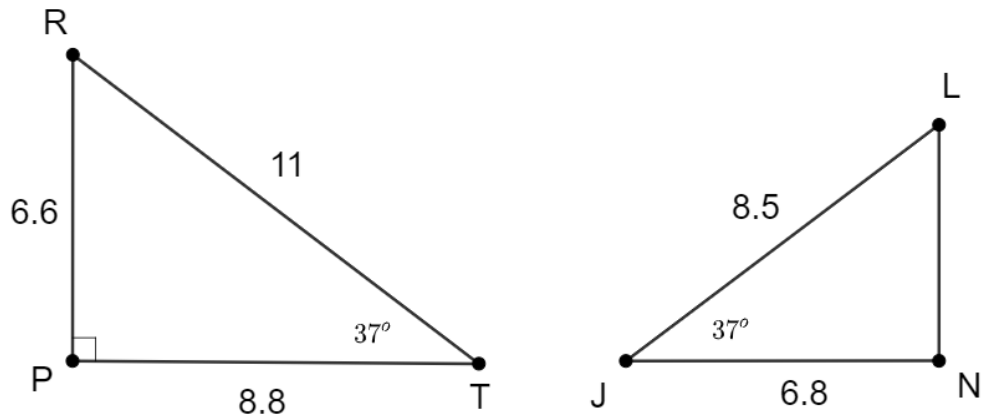
5. If $WZ = 3$, $XY = 10.5$, and $AD = 2$, determine $\overline{BC}$.

$$\frac{AD}{WZ} = \frac{BC}{XY}$$

$$\frac{2}{3} = \frac{BC}{10.5}$$

$$BC = \frac{(2)(10.5)}{3} = 7$$

$$BC = 7$$



$\triangle PRT$ and $\triangle JLN$ are shown. $PR = 6.6$, $PT = 8.8$, $RT = 11$, $JL = 8.5$, $JN = 6.8$, $\overline{RP} \perp \overline{PT}$, $m\angle PTR = 37^\circ$, and $m\angle LJN = 37^\circ$.

6. What similarity theorem or postulate could be used to prove the triangles are similar?

SAS ~

7. Write a similarity statement for $\triangle PRT$ and $\triangle JLN$.

$\triangle TPR \sim \triangle JNL$

Complete the table.

	Angle Measures	Side Proportion or Length
8.	$m\angle PRT = 53^\circ$	$\frac{RT}{LJ} = \frac{11}{8.5} = \frac{22}{17}$
9.	$m\angle JLN = 53^\circ$	$\frac{PT}{JN} = \frac{8.8}{6.8} = \frac{22}{17}$
10.	$m\angle LNJ = 90^\circ$	$\frac{11}{6.6} = \frac{8.5}{LN}$ $LN = 5.1$

Geometry
Unit 6
Lesson 11 Practice

Name Answer Key
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$DKLM$ and $PTRS$ are shown, where $DKLM \sim PTRS$ with $MD = 21$, $DK = 4y - 2$, $KL = 6x + 3$, $LM = 36$, $PT = 10$, $TR = 11$, and $RS = 12$.

1. Determine the value of each variable.

$$x = 3$$

$$y = 4$$

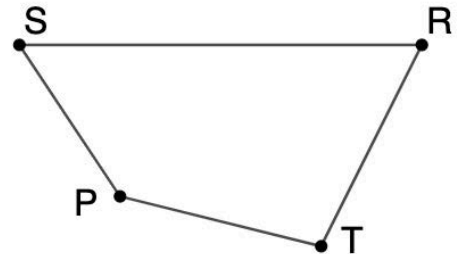
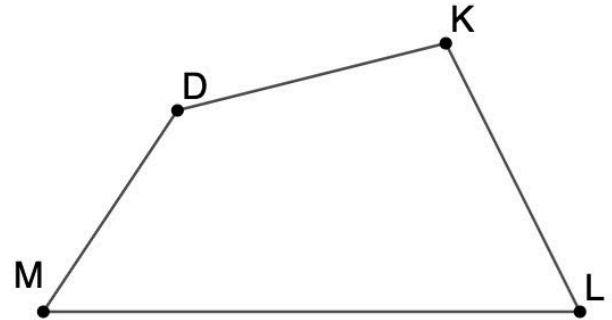
2. Determine the value of each side.

$$DK = 30$$

$$SP = 8$$

$$KL = 10$$

3. What is the perimeter of $DKLM$?



Consider additional measurements of angles for $DKLM$ and $PTRS$ where $m\angle MDK = (12x + 15)^\circ$, $m\angle SPT = (14x - 5)^\circ$, $m\angle DKL = 115^\circ$, $m\angle DML = (8y + 12)^\circ$, $m\angle PSR = (13y - 18)^\circ$.

4. Determine the value of each variable.

$$x = 10$$

$$y = 6$$

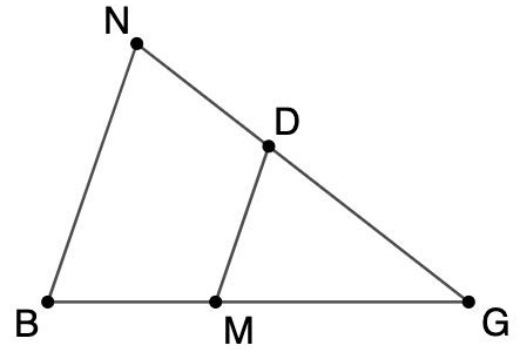
5. Determine the value of each angle.

$$m\angle MDK = 135^\circ$$

$$m\angle PTR = 115^\circ$$

$$m\angle PSR = 60^\circ$$

$\triangle NBG$ and $\triangle DMG$ are shown, where $\triangle NBG \sim \triangle DMG$ with $DM = 6$, $NB = 21$, $DG = 4$, $MG = 2x + 3$, and $BG = 13x - 1.5$.



6. What is the scale factor from $\triangle DMG$ to $\triangle NBG$?

$$\frac{21}{6} = 3.5$$

7. What is the value of the variable, x ?

$$3.5(2x + 3) = 13x - 1.5$$

$$x = 2$$

8. What is the length of BG ?

$$13(2) - 1.5 = 24.5 = BG$$

Consider additional measurements of angles for $\triangle NBG$ and $\triangle DMG$ where $m\angle GNB = (20y + 10)^\circ$, $m\angle GMD = (5x - 5)^\circ$, $m\angle GBN = (4x + 10)^\circ$, $m\angle GDM = (25y - 5)^\circ$.

9. Determine the value of each variable.

$$x = 15$$

$$y = 3$$

10. Determine the value of each angle.

$$m\angle GNB = 70^\circ$$

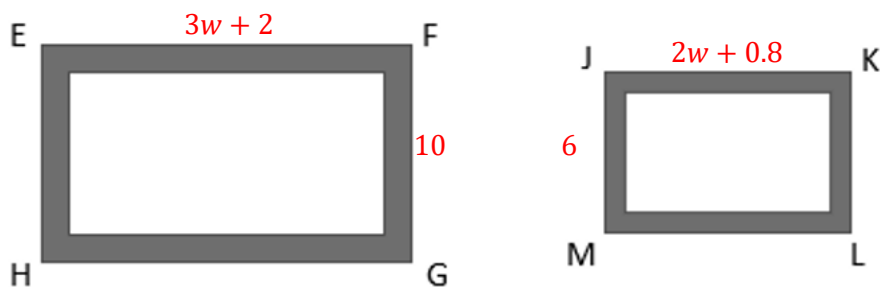
$$m\angle GBN = 70^\circ$$

$$m\angle DGM = 40^\circ$$

Geometry
Unit 6
Lesson 12 Practice

Name Answer Key
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Tamika is custom building frames for a set of family photos. She has designed two similar rectangular frames, $EFGH$ and $JKLM$. The diagram shows the two different sized frames, where $JM = 6$ inches (in.), $EF = (3w + 2)$ in., $FG = 10$ in., and $JK = (2w + 0.8)$ in.



1. What is the value of w ?

$$\frac{2w+0.8}{3w+2} = \frac{6}{10}$$

$$10(2w + 0.8) = 6(3w + 2)$$

$$2w = 4$$

$$20w + 8 = 18w + 12$$

$$w = 2$$

2. What is the area of $EFGH$? Include units.

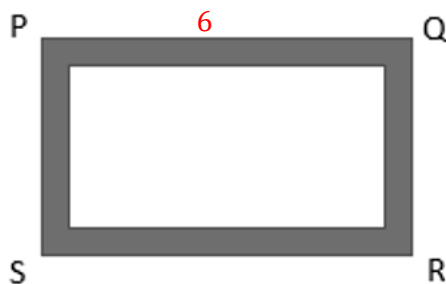
$$A = EF \cdot FG = [3(2) + 2] \cdot 10 = 80 \text{ in}^2$$

3. What is the perimeter of $JKLM$? Include units.

$$JK = 2(2) + 0.8 = 4.8 \text{ in}$$

$$P = 2 \cdot 4.8 + 2 \cdot 6 = 21.6 \text{ in}$$

Tamika created a third similar rectangular frame, $PQRS$, shown below, where $PQ = 6$ in.



4. What is the length of $\overline{SP}$? Include units.

$$SP = 7.5 \text{ in}$$

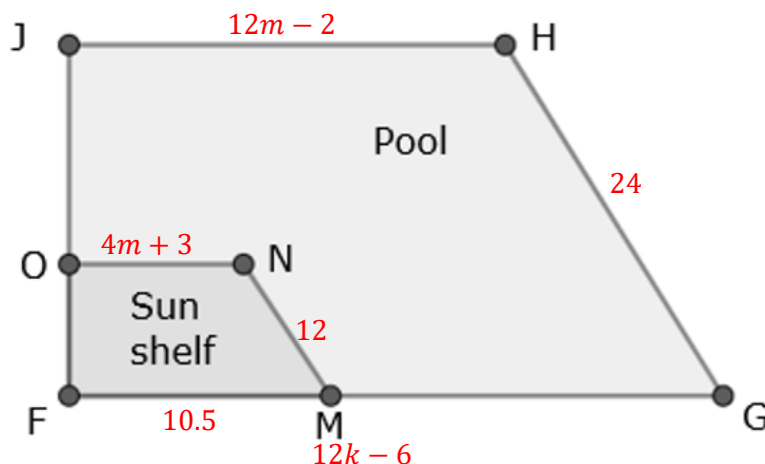
$$\frac{6}{8} = \frac{SP}{10}$$

$$SP = \frac{(10)(6)}{8} = 7.5 \text{ in}$$

5. What is the perimeter, in inches, of $PQRS$?

$$P = 2 \cdot 6 + 2 \cdot 7.5 = 27 \text{ in}$$

Shelley and her family are having a pool built in the backyard. The pool, $FGHJ$, is shaped like a trapezoid, with a smaller similar trapezoid, $FMNO$, for a sun shelf. A diagram of the pool is shown, where $FG = (12k - 6)$ feet (ft), $FM = 10.5$ ft, $NM = 12$ ft, $HG = 24$ ft, $HJ = (12m - 2)$ ft, and $NO = (4m + 3)$ ft.



6. Determine the value of each variable:

$$k = \underline{2.25} \quad \frac{10.5}{12k-6} = \frac{12}{24} \quad 144k = 324 \quad \left| \quad \frac{4m+3}{12m-2} = \frac{12}{24} \quad 2m = 4 \right.$$

$$m = \underline{2} \quad (24)(10.5) = 12(12k - 6) \quad k = 2.25 \quad \left| \quad 4m + 3 = \frac{1}{2}(12m - 2) \quad m = 2 \right.$$

$$252 = 144k - 72 \quad \left| \quad 4m + 3 = 6m - 1 \right.$$

7. Determine the value of each side:

$$HJ = \underline{22} \quad HJ = 12m - 2 = 12(2) - 2 = 22$$

$$FG = \underline{21} \quad FG = 12k - 6 = 12(2.25) - 6 = 21$$

8. If $HJ = JF$, what is the perimeter, in feet, of the pool?

$$P = HJ + JF + FG + HG = 22 + 22 + 21 + 24 = \boxed{89 \text{ ft}}$$

Some measurements of angles of $FGHJ$ and $FMNO$ are given.

$$m\angle FMN = (15y + 10)^\circ, m\angle FGH = (18y - 2)^\circ, m\angle OFM = 90^\circ,$$

$$m\angle FJH = (5w + 15)^\circ, m\angle FON = (15w - 15)^\circ$$

9. Determine the value of each variable:

$$y = \underline{4} \quad 15y + 10 = 18y - 2 \quad 5w + 15 = 15w - 15$$

$$w = \underline{3} \quad 3y = 12 \quad 10w = 30$$

$$y = 4 \quad w = 3$$

10. What is the measure of $\angle ONM$?

$$m\angle OFM + m\angle FON + m\angle ONM + m\angle FMN = 360^\circ$$

$$\boxed{m\angle ONM = 170^\circ}$$

$$90^\circ + [(15)(3) - 15]^\circ + m\angle ONM + [(15)(4) + 10]^\circ = 360^\circ$$

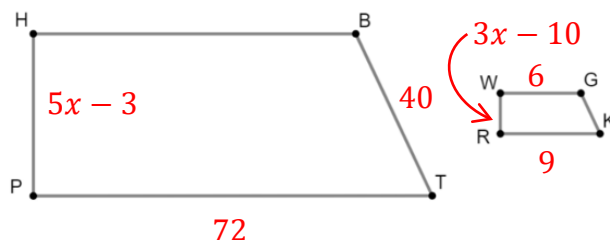
$$90^\circ + 30^\circ + m\angle ONM + 70^\circ = 360^\circ$$

Geometry
Unit 6
Lesson 13 Practice

Name Answer Key
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Wendy, a party planner, is designing a banner for a retirement party. Her client wants two similar banners: a large banner, $HBTP$, for the main entrance, and a similar sized banner, $WGKR$, for a photo booth.

Wendy knows that $PT = 72$ inches (in.),
 $HP = (5x - 3)$ in., $WG = 6$ in., $WR = (3x - 10)$ in.,
 $BT = 40$ in., and $RK = 9$ in.



1. What scale factor will map $WGKR$ to $HBTP$?

$$\frac{72}{9} = 8$$

2. What is the perimeter of $HBTP$, the main entrance banner?

$$HB = 6 \cdot 8 = 48 \text{ in}$$

$$HP = 5 \left(\frac{77}{19} \right) - 3 \approx 17.26 \text{ in}$$

$$P = 72 + 40 + 48 + 17.26 = 177.26 \text{ in}$$

$$5x - 3 = 8(3x - 10)$$

$$5x - 3 = 24x - 80$$

$$19x = 77$$

$$x = \frac{77}{19} \approx 4.0526$$

3. What is the perimeter of $WGKR$, the photo booth banner?

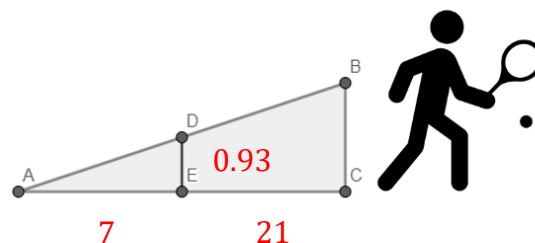
$$GK = \frac{40}{8} = 5$$

$$WR = 3 \left(\frac{77}{19} \right) - 10 \approx 2.16 \text{ in}$$

$$P = 9 + 5 + 6 + 2.16 = 22.16 \text{ in}$$

Answers to #2 and #3 may vary slightly due to rounding of the value of x .

In the diagram, $\triangle ADE \sim \triangle ABC$, $AE = 7$ meters (m),
 $DE = 0.93$ m, and $EC = 21$ m. Serena is doing a tennis drill where she wants to hit the ball so that it will just pass over the net, $\overline{DE}$, and land 7 m away from the base of the net.



4. What scale factor will map $\triangle ABC$ to $\triangle ADE$?

$$\frac{7}{28} = \frac{1}{4}$$

5. What is the height, represented by $\overline{BC}$, of the tennis racket?

$$\frac{7}{28} = \frac{0.93}{BC}$$

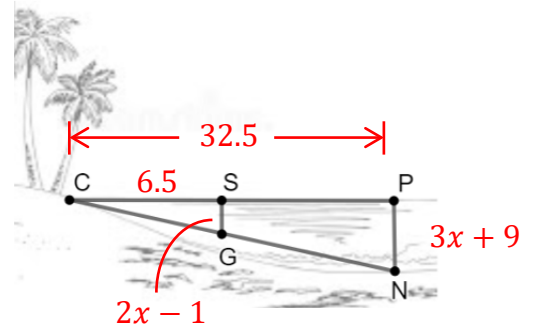
$$7 \cdot BC = (0.93)(28)$$

$$BC = 3.72 \text{ m}$$

A lake front property owner is building a dock which extends the length of $\overline{CP}$ into the lake. $\overline{PN}$ and $\overline{SG}$ are the two main support posts for the dock.

$\triangle CSG \sim \triangle CPN$, $CS = 6.5$ feet (ft), $SG = (2x - 1)$ ft,

$PN = (3x + 9)$ ft, and $CP = 32.5$ ft.



6. What scale factor will map $\triangle CPN$ to $\triangle CSG$?

$$\frac{6.5}{32.5} = 0.2 \text{ or } \frac{1}{5}$$

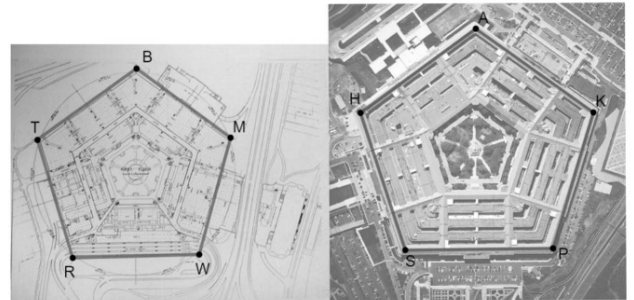
7. What is the depth of the longer support post, represented by $\overline{PN}$?

$$\begin{array}{lcl} \frac{2x - 1}{3x + 9} = \frac{1}{5} & 5(2x - 1) = 3x + 9 & PN = 3x + 9 \\ 10x - 5 = 3x + 9 & 7x = 14 \rightarrow x = 2 & PN = 3(2) + 9 \\ & & PN = 15 \text{ ft} \end{array}$$

8. What is the depth of the shorter support post, represented by $\overline{SG}$?

$$\begin{array}{l} SG = 2x - 1 \\ SG = 2(2) - 1 \\ SG = 3 \text{ ft} \end{array}$$

Cesar wants to create a scale model of the Pentagon, where all sides are equal, for his social studies project. A copy of the blueprint, $BMWRT$, will guide him in building a similar scale model, $AKPSH$. The blueprint's sides, $BMWRT$, are $\frac{1}{6}$ of the scale model Pentagon, $AKPSH$. $TB = (0.5x + 5)$ centimeters (cm) and $HA = (5x + 10)$ cm.



9. What is the perimeter of the Pentagon in the blueprint, $BMWRT$?

$$\begin{array}{lcl} \frac{0.5x + 5}{5x + 10} = \frac{1}{6} & 6(0.5x + 5) = 5x + 10 & P = 5 \cdot [(0.5)(10) + 5] \\ 3x + 30 = 5x + 10 & 2x = 20 & P = 50 \text{ cm} \\ x = 10 & & \end{array}$$

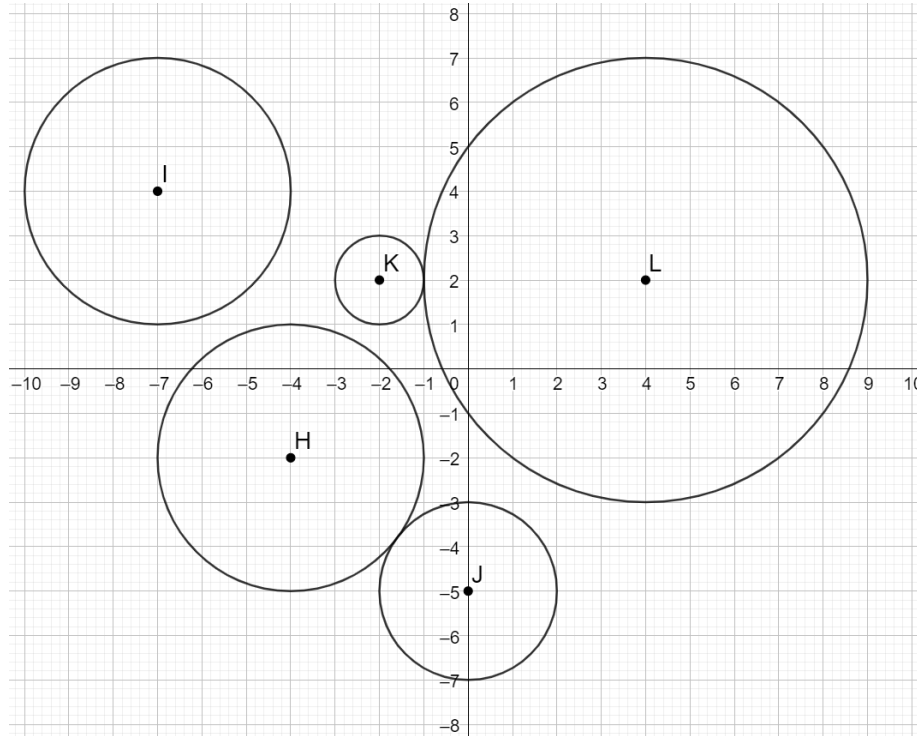
10. What is the perimeter of the scale model Pentagon, $AKPSH$?

$$\begin{array}{l} P = 5 \cdot [(5)(10) + 10] \\ P = 300 \text{ cm} \end{array}$$

Geometry
Unit 6
Lesson H1 Practice

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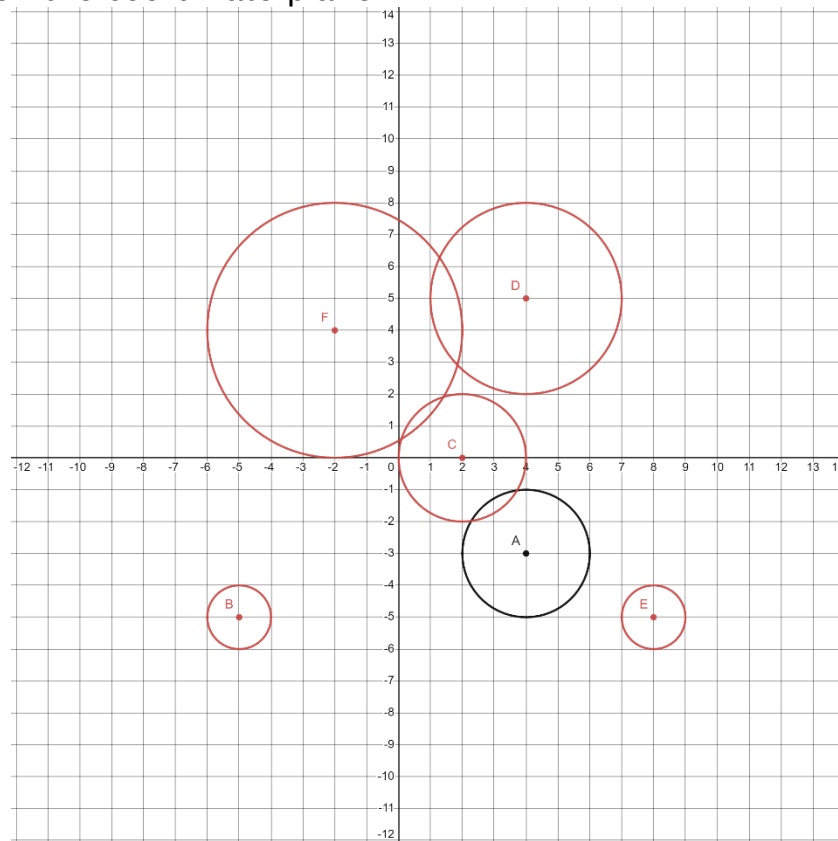
1. All circles are congruent
✗ similar.



Given Circle H as the preimage:

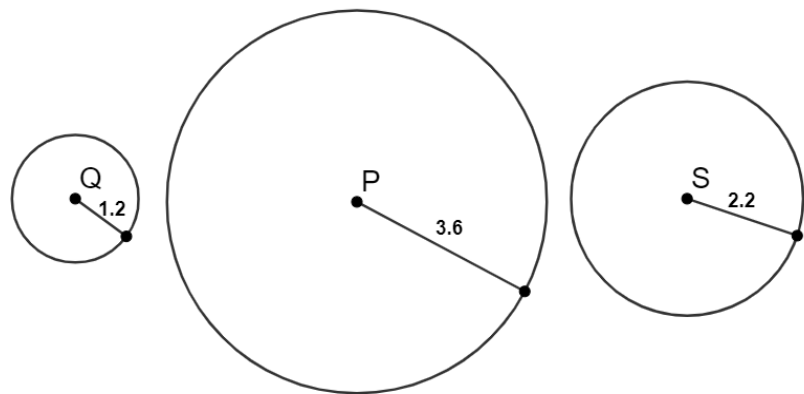
2. How can Circle H be mapped onto Circle I ?
Translation left 3, up 6.
3. How can Circle H be mapped onto Circle J ?
Translation right 4, down 3, then dilated by $\frac{2}{3}$.
4. How can Circle H be mapped onto Circle K ?
Translation right 2, up 4, then dilated by $\frac{1}{3}$.
5. How can Circle H be mapped onto Circle L ?
Translation right 8, up 4, then dilated by $\frac{5}{3}$.

Circle A is shown on the coordinate plane.



6. Draw Circle B , the result of translation 9 units left and 2 units down and a dilation of $\frac{1}{2}$ about the center of Circle A .
7. Draw Circle C , the result of translation 2 units left and 3 units up.
8. Draw Circle D , the result of a dilation of $\frac{3}{2}$ about the center of Circle A and a translation 8 units up.
9. Draw Circle E , the result of translation 4 units right and 2 units down and a dilation of $\frac{1}{2}$ about the center of Circle A .
10. Draw Circle F , the result of translation 6 units left and 7 units up and a dilation of 2 about the center of Circle A .

Circles Q , P , and S are shown. The radius of Circle Q is 1.2 units long. The radius of Circle P is 3.6 units long. The radius of Circle S is 2.2 units long.



1. Complete the following table. Round to the nearest thousandth. Use the π button on your calculator.

Circle	Radius	Circumference
Q	1.2	7.540
P	3.6	22.619
S	2.2	13.823

2. Complete the table.

Circle Q and P	
Ratio of Radius of Circle Q and Radius of Circle P	$\frac{1}{3}$
Ratio of Circumference of Circle Q and Circumference of Circle P	$\frac{1}{3}$

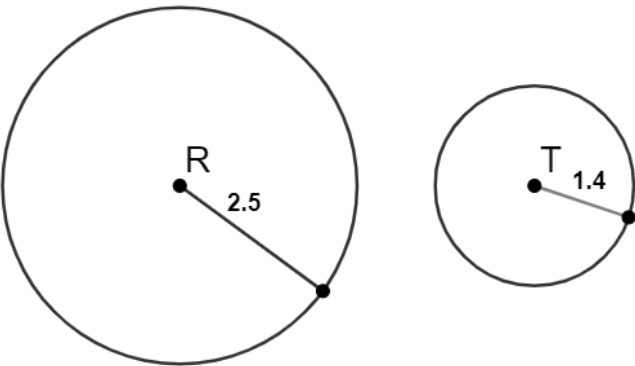
3. Complete the table.

Circle P and S	
Ratio of Radius of Circle P to Radius of Circle S	1.64
Ratio of Circumference of Circle P to Circumference of Circle S	1.64

4. Complete the table.

Circle Q and S	
Ratio of Radius of Circle Q to Radius of Circle S	0.545
Ratio of Circumference of Circle Q to Circumference of Circle S	0.545

Circles R and T are shown. The radius of Circle R is 2.5 units long. The radius of Circle T is 1.4 units long.



Show Circle R is similar to Circle T by completing the proof by contradiction:

5. Assume that Circle R is not similar to Circle T .
6. The ratio of the radius of Circle R to the radius of Circle T is 1.79.
7. The ratio of the circumference of Circle R to the circumference of Circle T is 1.79.
8. The ratio of the radius and the ratio of the circumference are equal, so corresponding parts of Circle R and Circle T are proportional.
9. Since the corresponding parts of Circle R and Circle T are proportional, the circles are similar. This is a contradiction, so the assumption that Circle R is not similar to Circle T is not true.
10. All circles are

o congruent

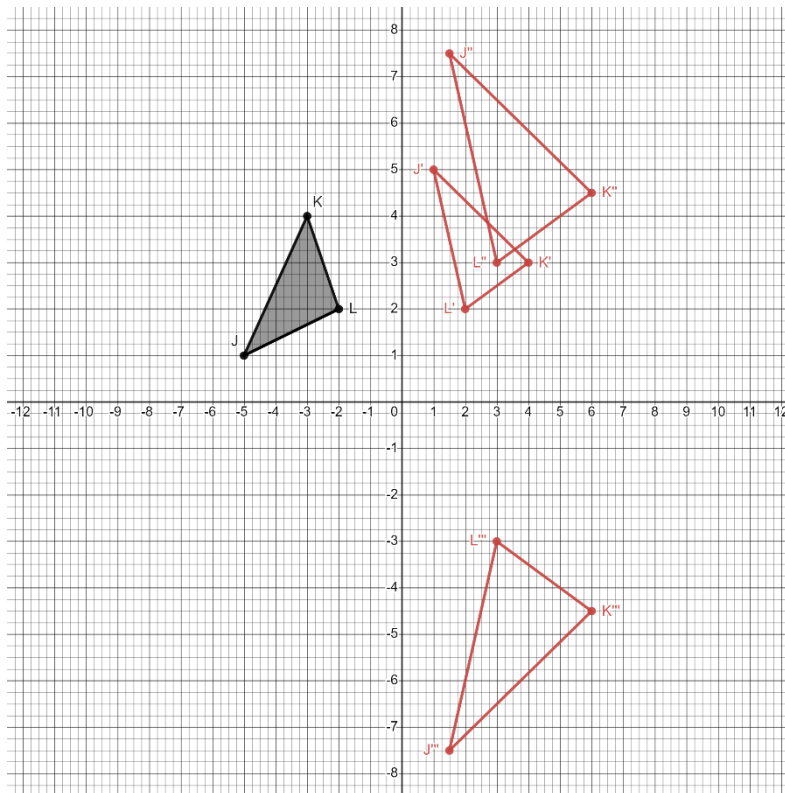
✗ similar

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Geometry
Unit 6
Lesson H3 Practice

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$\triangle JKL$ is shown.



Draw the result of each transformation in the following sequence:

- $\triangle J'K'L'$ as a result of $(x, y) \rightarrow (y, -x)$
- $\triangle J''K''L''$ as a result of $(x, y) \rightarrow \left(\frac{3}{2}x, \frac{3}{2}y\right)$
- $\triangle J'''K'''L'''$ as a result of $(x, y) \rightarrow (x, -y)$

Indicate the correct relationship for the resulting image:

- Image $\triangle J'K'L'$ is

☒ congruent
☐ similar

 to preimage $\triangle JKL$.
- Image $\triangle J''K''L''$ is

☐ congruent
☒ similar

 to preimage $\triangle JKL$.
- Image $\triangle J'''K'''L'''$ is

☐ congruent
☒ similar

 to preimage $\triangle JKL$.

$\triangle ABC$ and $\triangle DEF$ are shown.

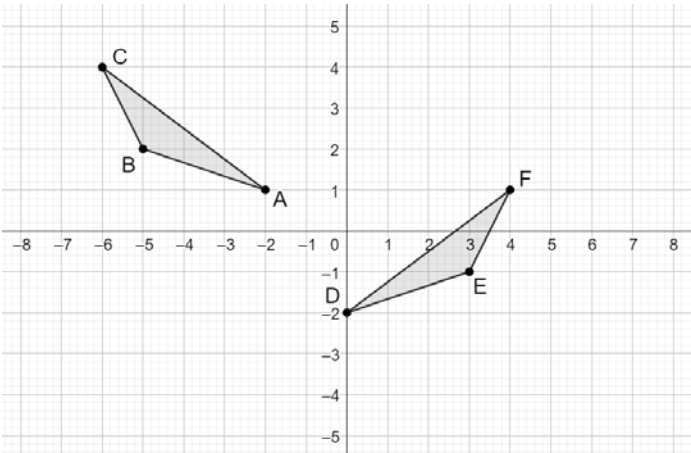
Assume: $\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF}$

7. Write a series of transformations that will map $\triangle ABC$ onto $\triangle DEF$ so that $\triangle ABD \sim \triangle DEF$.

Answers may vary.

Example:

$(x,y) \rightarrow (-x,y)$ then $(x,y) \rightarrow (x-2,y-3)$



8. If it is given that $\triangle DEF$ is a result of $(x,y) \rightarrow (-x,y)$ then $(x,y) \rightarrow (x-2,y-3)$, then the triangles are

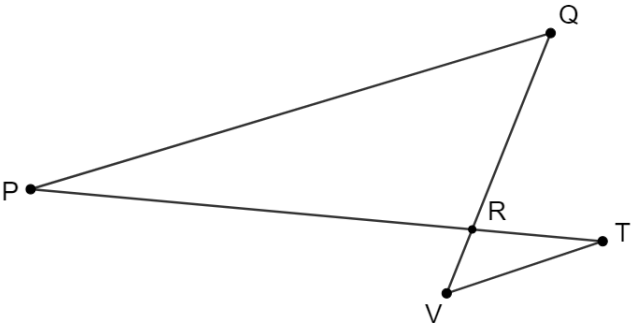
☒ congruent

☐ similar

.

$\triangle PQR$ and $\triangle TVR$ are shown.

Complete the table:



	If Given	True or False? $\triangle PQR \sim \triangle TVR$ by AA Similarity	Justification
9.	$\frac{TR}{PR} = \frac{VR}{QR}$	☐ True ☒ False	Triangles are similar by SAS similarity. Two sets of angles are not proven congruent.
10.	$\overline{PQ} \parallel \overline{VT}$	☒ True ☐ False	$\angle PQR \cong \angle TVR$ by alternate interior angles. $\angle QPR \cong \angle VTR$ by alternate interior angles. $\angle PRQ \cong \angle TRV$ by vertical angles. *Any combination of two of these three.